# 0002. Add Two Numbers

Medium · Linked List

leetcode.com/problems/add-two-numbers

# Problem

- Two non-negative numbers are stored as reversed digit lists. Return their sum as a list in the same format.
- Example: 2→4→3 (342) + 5→6→4 (465) = 7→0→8 (807).

# Intuition

- Reversed order means the head is the ones digit, exactly the order of grade-school addition.
- Add digit by digit with a carry, and continue while either list or the carry remains.

# Brute Force

- Convert to integers and add: overflows for long lists (up to 100 digits).

# Optimal Approach

- Dummy, carry = 0.
- While a or b or carry: sum = carry + a.val? + b.val?; append sum % 10; carry = sum / 10.

# Worked Example

```
9->9 + 1.

9+1+0 = 10 -> digit 0 carry 1
9+0+1 = 10 -> digit 0 carry 1
carry 1    -> digit 1
result 0->0->1 (99 + 1 = 100)
```

# Java Solution

```java
public class AddTwoNumbers {
    public ListNode addTwoNumbers(ListNode a, ListNode b) {
        ListNode dummy = new ListNode(0), tail = dummy;
        int carry = 0;
        while (a != null || b != null || carry != 0) {
            int sum = carry;
            if (a != null) { sum += a.val; a = a.next; }
            if (b != null) { sum += b.val; b = b.next; }
            tail.next = new ListNode(sum % 10);
            tail = tail.next;
            carry = sum / 10;
        }
        return dummy.next;
    }
}
```

# Complexity

- Time: O(max(m, n)). Space: O(max(m, n)) for the output.

# Pitfalls

- The final carry creates an extra node.